جامعة حفر الباطن
University of Hafr Al Batin

Lecture 2: Internet History, Administration and Infrastructure

Objectives

- History of the Internet
- Overview of The Internet today
- The Internet Architecture and Administration.

History of the Internet

- Before 1960, there are communication networks such as telegraph and telephone network
- Computer networks should be able to handle bursty data, which means data received at variable rate
- Theory of packet switching for bursty traffic was first presented in 1961 by Leonard Kleinrock
- In 1967, during ACM meeting Advanced Research Project Agency presented **ARPANAT**.
  - By 1969 **ARPANET** was a reality

Birth of the Internet

- In 1972, Vint Cerf and Bob Kahn, being part of ARPANET collaborated on what they called internetting project.
  - They devised the idea of gateway to serve intermediate hardware to transfer data from one network to another
- In 1973, Cerf and Kahn presented TCP protocol to achieve end-to-end delivery of packets
- By 1977, an internet consisting of three networks was demonstrated
  - ARPANET, packet radio and packet satellite was successfully tested.

Birth of the Internet

- Shortly thereafter, the authorities made a decision to split TCP into two protocols TCP/IP
  - IP to handle datagram routing
  - TCP to cater for higher level functions such as segmentation, reassembly and error detection
- In 1983, authorities abolished the original ARPANET.
  - and TCP/IP becomes the **official protocol for the ARPANET**.

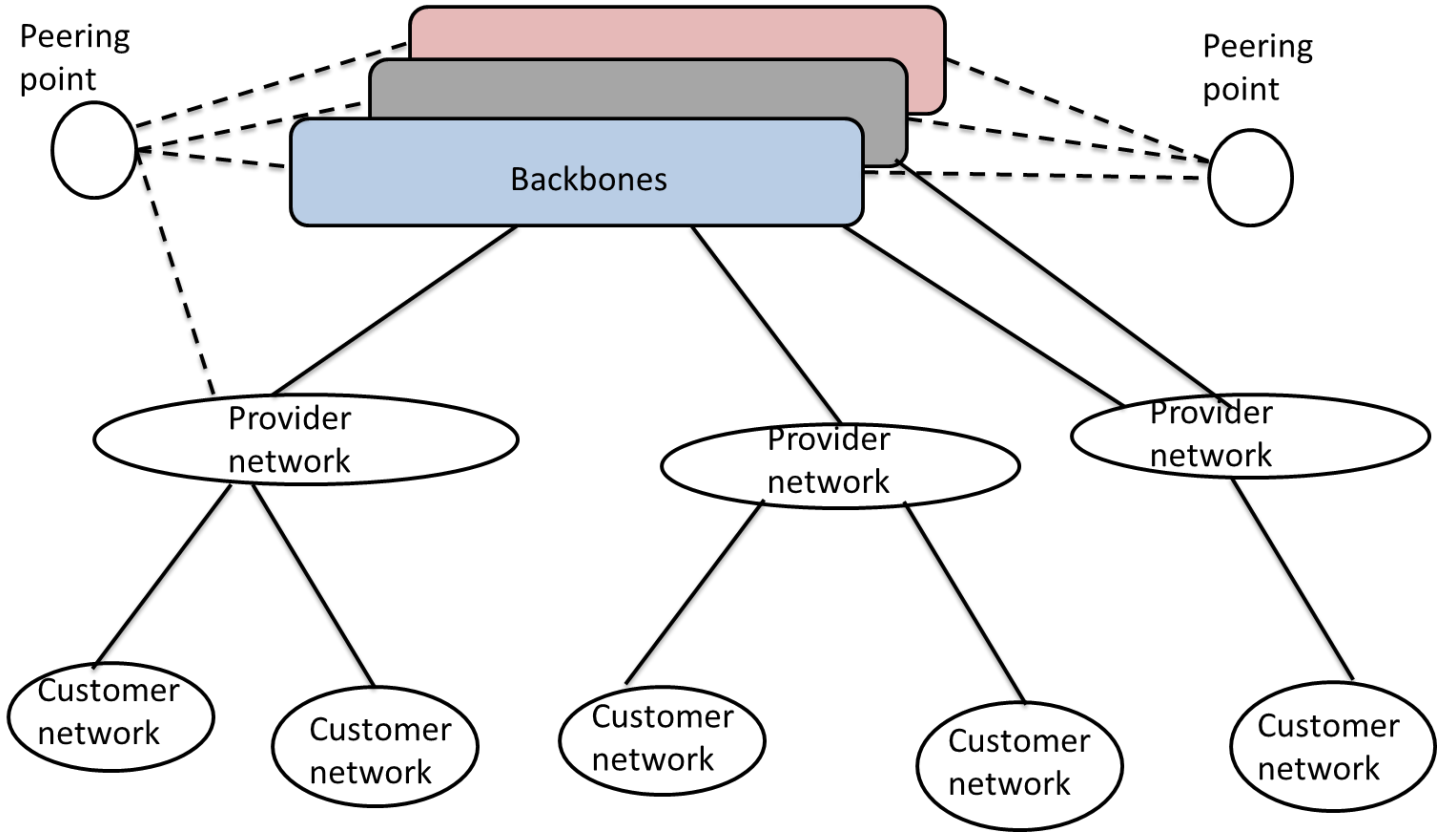
The Internet Today

- The internet is so popular nowadays due to the invention of new applications.
- In 1990, Tim Berner-Lee invented the WWW (World Wide Web)
  - Multimedia
  - Voice over IP(Telephony)
  - Video over IP (Skype),
  - View sharing (Youtube)
  - Television over IP
  - Peer-to-Peer Applications

The Internet

- **An internet is two or more networks that can communicate with each other**
- The most notable internet is called Internet
- The Internet is made up of several **backbones**, **provider networks**, and **customer networks.**
- At the top level , the backbones are large networks.
  - Owned by some communication companies such as Sprint, Verizon and AT&T
  - The backbones are connected through some complex switching called peering points
- At the second level, there are smaller networks called provider networks
  - Uses the services of backbones for a fee
- The customer networks are networks at the
  - uses the services provided by the provider network.
- **The backbones and provider networks are called Internet Service Providers (ISP)**

The Internet Architecture



Accessing the Internet

- Any user can be part of the Internet.
  - But the user must be physically connected to an ISP.
- The physical connection is normally done with point-to-point WAN

1. Using **Telephone Network**
   - **Dial Up** Service: Use of modem to switch between data and voice.
   - **DSL** service: line simultaneously used for voice and data communication.
2. Using **Cable Networks.**
3. Using **Wireless Networks.**
4. **Direct Connection** to the Internet: A large university with several campus can have an internetwork and directly connect to a regional ISP.

Internet Standards

- **Internet standard:** a thoroughly tested specification that is useful to and adhered by those who worked with the Internet.
- It is a formalized regulation that must be followed
- There is a strict procedure by which a specification attain Internet standard level:
  1. An Internet draft is a working document with no official status and has a six month life time
  2. Upon recommendation from the Internet authorities, a draft may be published as Request for Comment (RFC)
  3. RFC is edited assigned a number and made available to all interested parties.

Internet Administration

- The Internet Society (ISOC) – is an international non profit organization formed in 1992 to provide support to the Internet Standard process.
  - ISOC maintains and support other bodies such as IAB, IETF, IRTF and IANA
- Internet Architecture Board (IAB) is a technical advisor of ISOC.
  - IAB accomplishes its tasks through its components Internet Engineering Task Force (IETF) and Internet Research Task Force (IRTF)
- IETF is managed by **Internet Engineering Steering Group (IESG)**
  - Responsible for identifying operational problems and proposing solution.
  - Develops and review specification intended for Standards.
- The Internet Research Task Force (IRTF) is a forum responsible Long term research towards Internet protocol, applications, architecture and technology.